SESSION 12 – Hypothesis Testing Concepts

Task 1. Null and Alternative Hypotheses

Question: State the null and alternative hypotheses for testing whether a new cashback offer on a food delivery app like Zomato has increased the average number of daily orders compared to before the offer.

Answer:

Let:

- μ_1 = Average daily orders before the cashback offer.
- μ_2 = Average daily orders after the cashback offer.

Null Hypothesis (H_0):

The cashback offer has not increased the average number of daily orders.

H0:μ2≤μ1H_0: \mu_2 \leq \mu_1H0:μ2≤μ1

Alternative Hypothesis (H_1):

The cashback offer has increased the average number of daily orders.

H1:μ2>μ1H_1: \mu_2 > \mu_1H1:μ2>μ1

Conclusion: We test whether the new cashback offer has a positive effect on the average daily orders.

Task 2. Type I or Type II Error

Question: A music streaming app claims its new playlist recommendation algorithm increases average listening time from 40 to 45 minutes. Which type of error occurs if you wrongly conclude the algorithm works when it actually doesn't?

Answer:

Type I Error (False Positive).

A Type I error occurs when we reject a true null hypothesis and conclude that the new recommendation algorithm increases listening time, even though it actually has no real improvement.

Task 3. Interpret the p-value

Question: Suppose you run an A/B test on a shopping app like Flipkart and get a p-value of 0.03 when testing if a new discount banner increases purchases. At a 95% confidence level, state whether you reject or fail to reject the null hypothesis and explain what this means.

Answer:

Given:

- P-value = 0.03
- Confidence level = 95%
- Significance level (α) = 1 − 0.95 = 0.05

Decision:

Since:

0.03<0.050.03 < 0.050.03<0.05

We reject the null hypothesis (H_0).

Interpretation:

There is statistically significant evidence that the new discount banner increases purchases on Flipkart. This suggests that the banner may be effective in encouraging customers to buy products.

Task 4. Explain the 99% confidence level

Question: For a cricket app's push notification feature, you want to be 99% confident that sending match alerts increases user engagement. Explain what the 99% confidence level means and how it affects your decision to accept or reject the feature.

Answer:

A 99% confidence level means that we use a significance level of 1% ($\alpha = 0.01$) for our hypothesis test.

It means we require stronger statistical evidence before concluding that match alerts increase user engagement.

- If the p-value is less than 0.01, we reject the null hypothesis.
- If the p-value is greater than or equal to 0.01, we fail to reject the null hypothesis.

Example:

If the p-value is 0.005, we reject the null hypothesis because 0.005 is less than 0.01. This provides statistically significant evidence that match alerts increase user engagement.

If the p-value is 0.03, we fail to reject the null hypothesis at the 99% confidence level.

Conclusion: A 99% confidence level requires stronger evidence before concluding that the new push notification feature increases user engagement.